Integration

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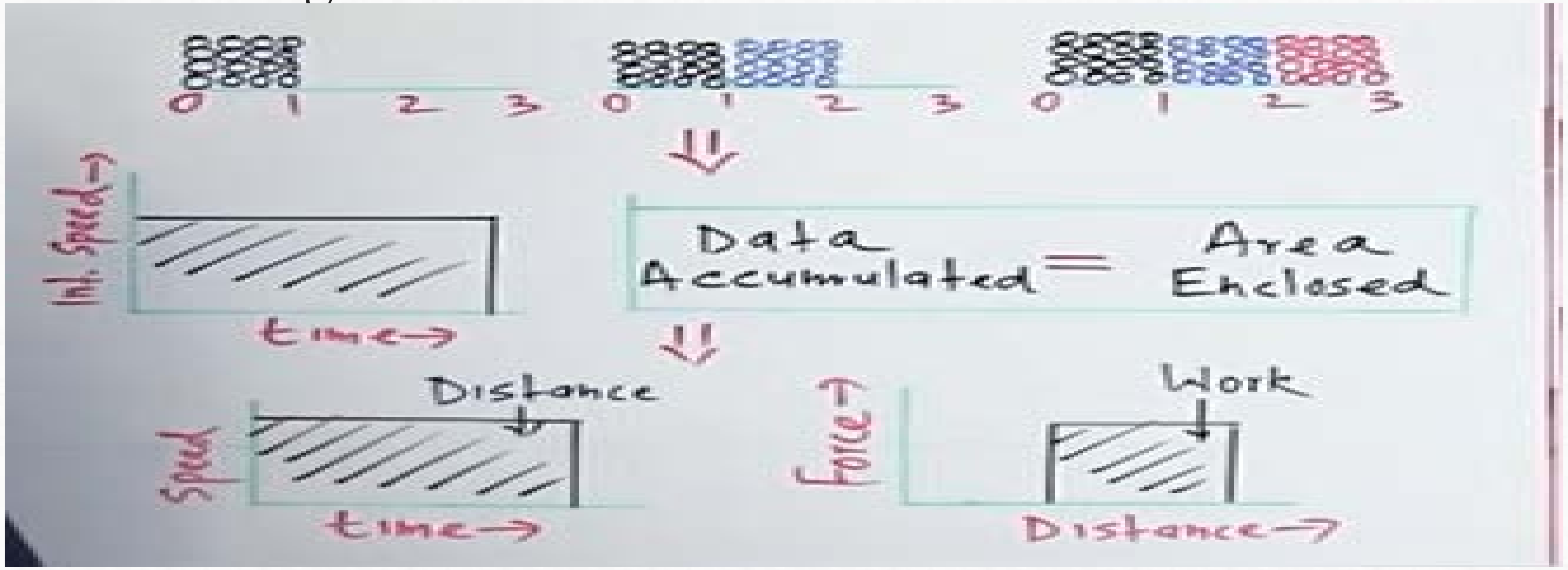
Contents

What is integration?

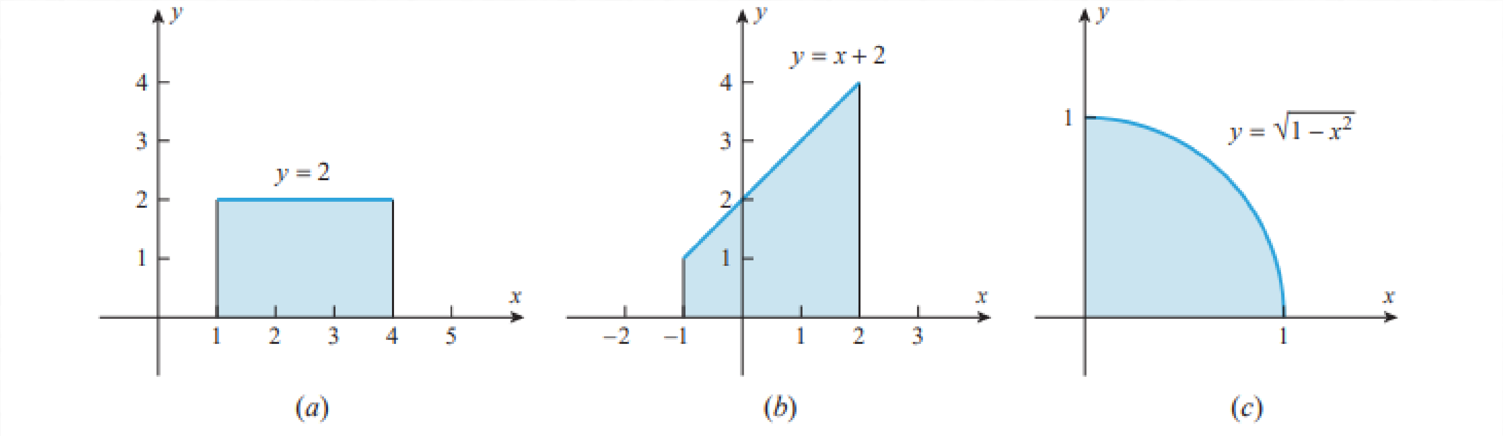
Definition: Integration is the process of finding the anti-derivative or the area under a curve .

What is integration?

- Integration tells us how much something is accumulating.
- Example: Downloading a movie.



Find area under the curves



Integral tells Area enclosed

- All cases so far area enclosed are in straight lines and hence easy to find.

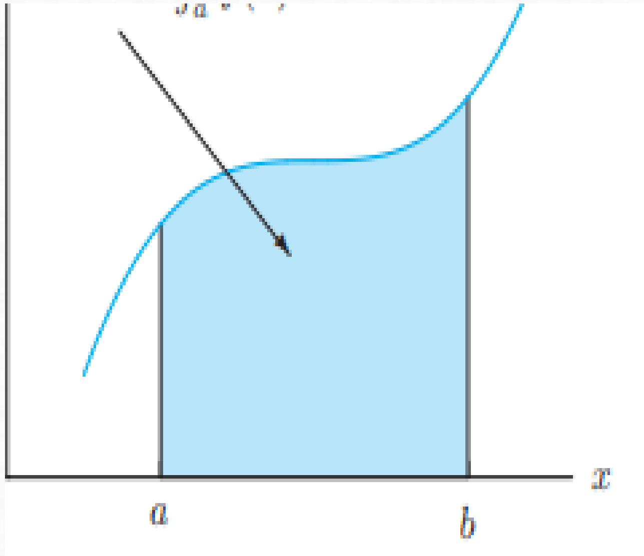
BUT

That rarely happened.

What is area under curve?

Integration as Summation

- All the above shapes were regular, but what about the area under the curve of figure given on the left.



Small rectangles strips are used to estimate the Area

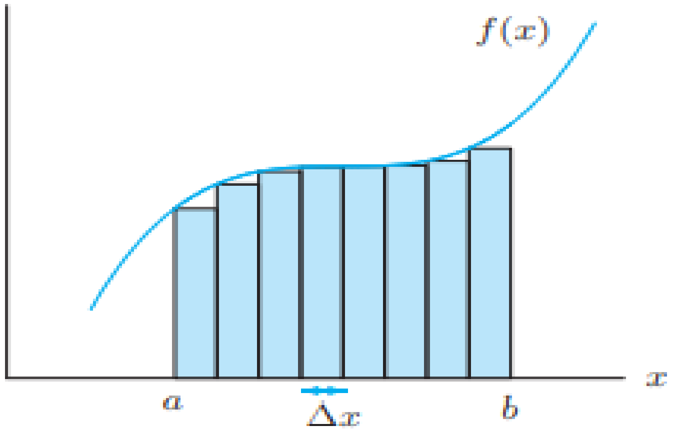


Figure 5.26: Area of rectangles approximating the area under the curve

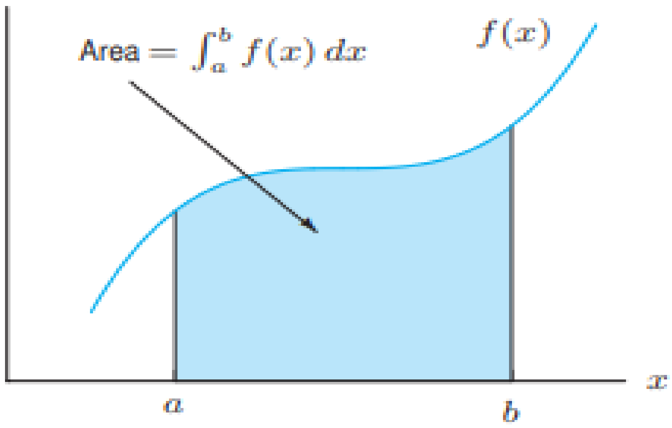


Figure 5.27: Shaded area is the definite integral $\int_a^b f(x)\,dx$

When $f(x)$ is positive and $a < b$:

$$\text{Area under graph of } f \text{ between } a \text{ and } b = \int_a^b f(x)\,dx.$$

More rectangles give more Accurate Area under the curve.

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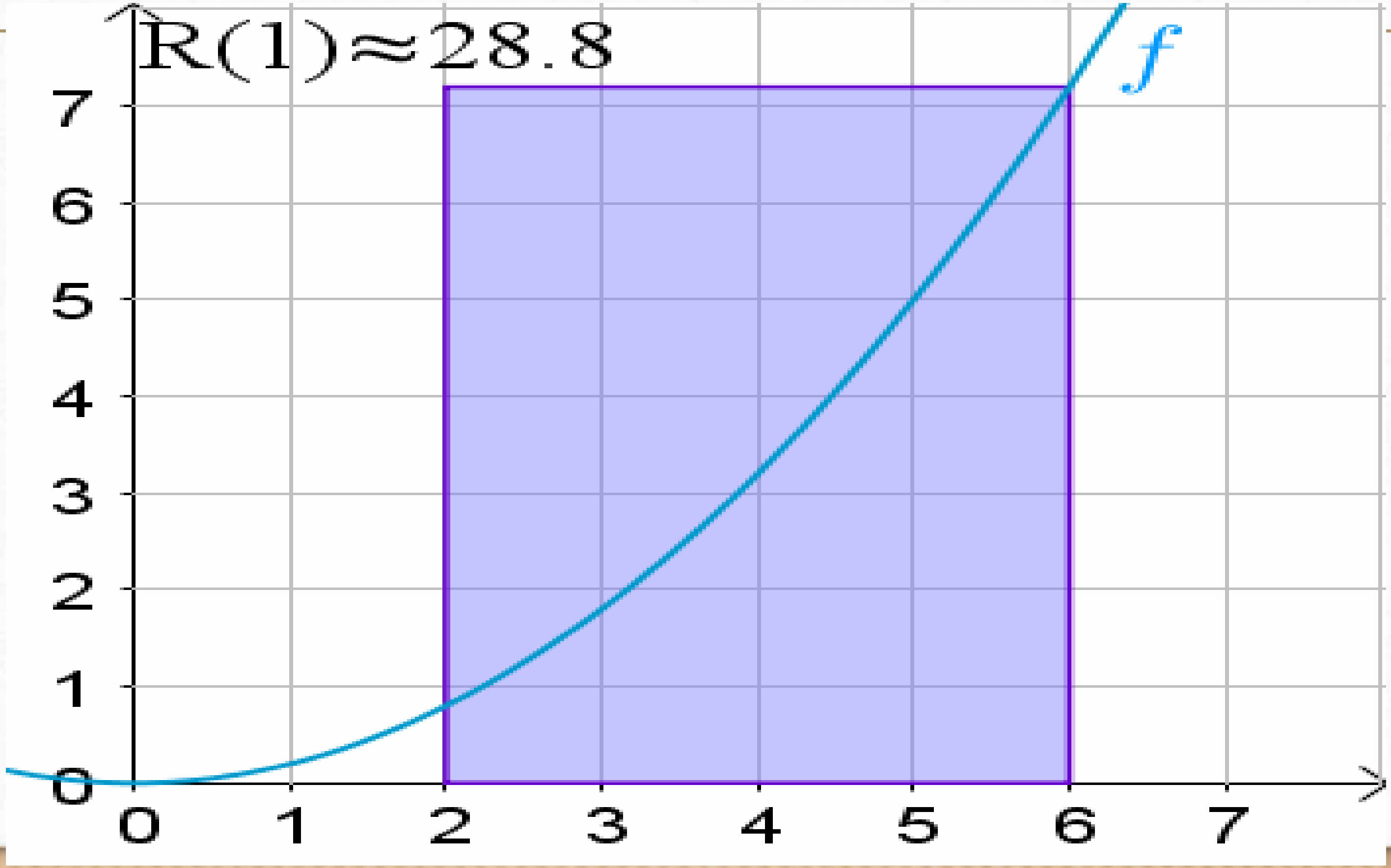
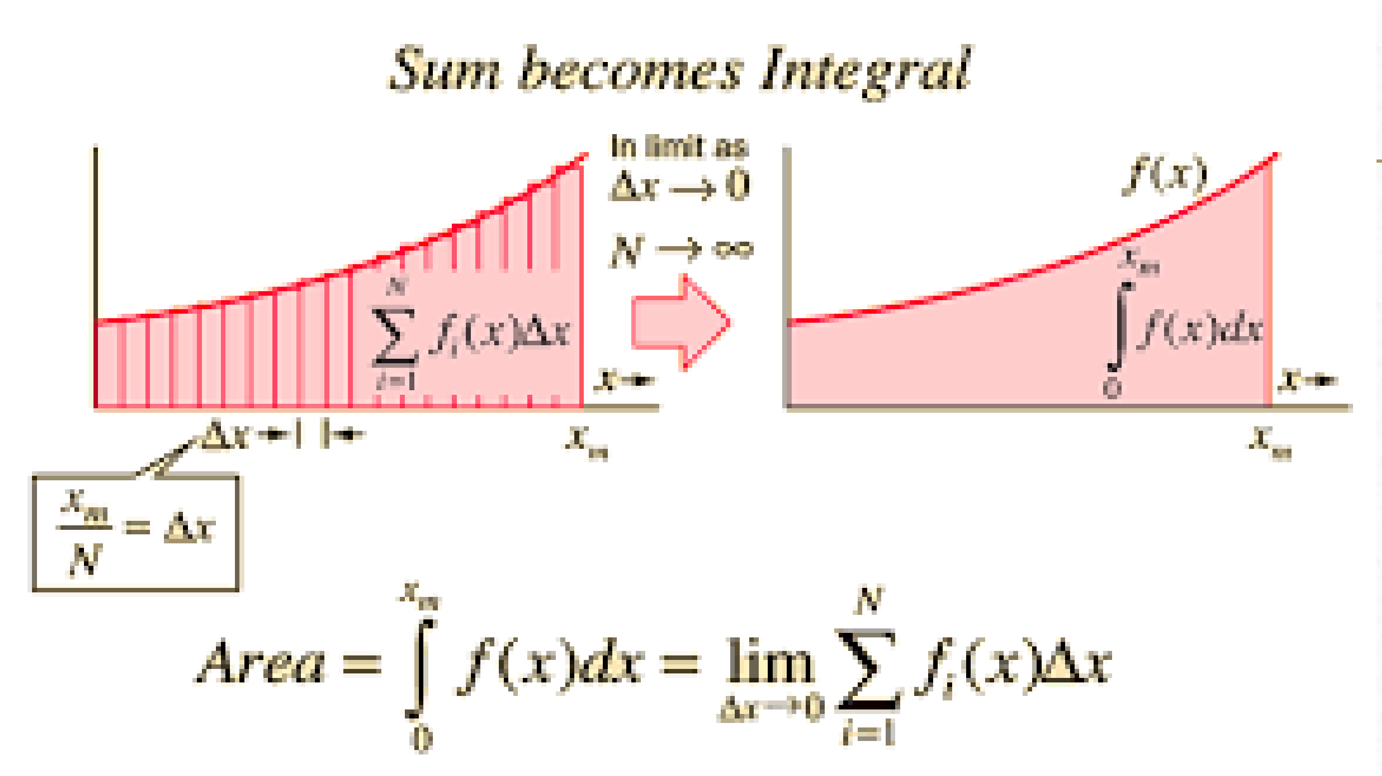


Integration as Sum of Infinite Rectangles

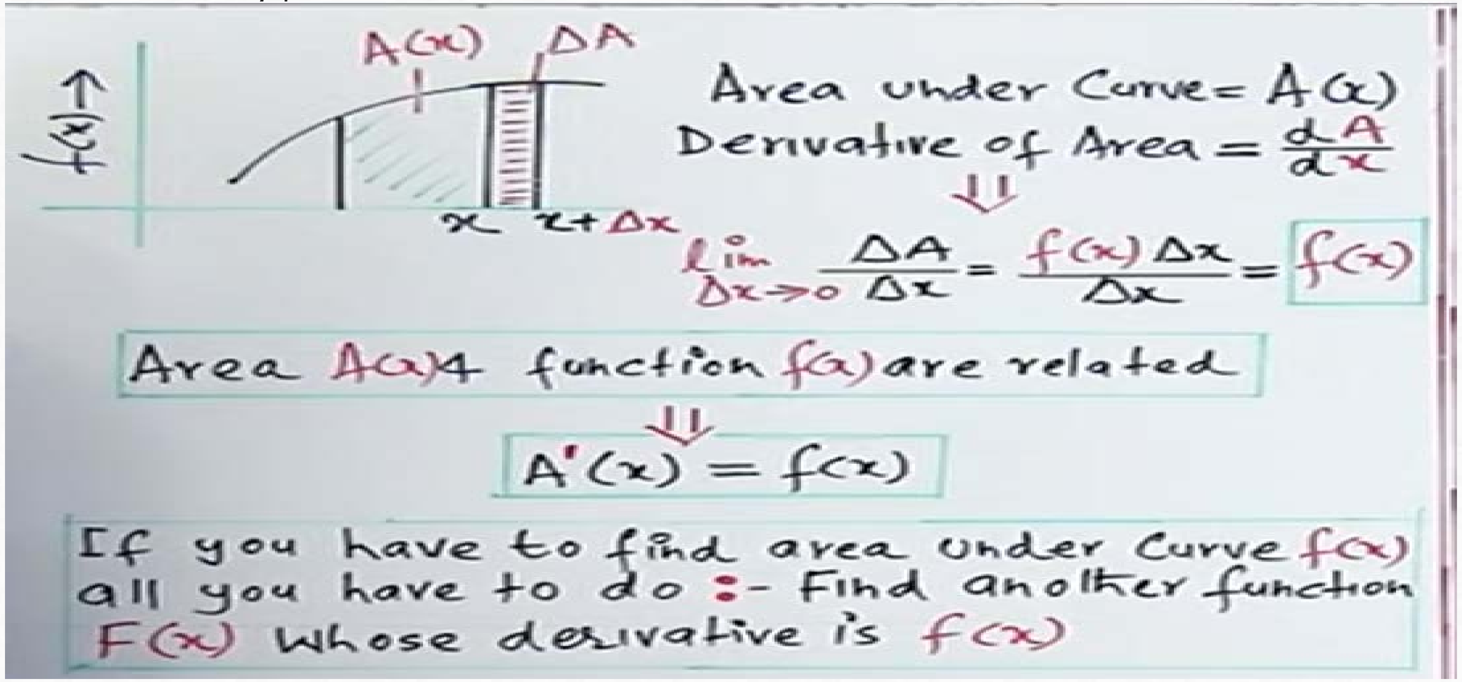


- *Do you know?*

The sign of integration, integral, $\int$ is actually an elongated S (“S” for SUM).

Integration as Anti-Derivative

- In 17th century we found a huge SHORTCUT that shortcut became the concept of integral



Integration as Anti-Derivative

- Function $f(x)$

Differentiation

Anti-Differentiation
or
Integration

- Slope function $f'(x)$
- Finding

Indefinite Integral

- The **indefinite integral** of is written as:

$$\int f(x)\,dx = F(x) + C,$$

∫: This is the integral symbol, indicating that we are integrating the function.

- **f(x)**: The function to be integrated.
- **dx**: This indicates the variable of integration.
- **F(x)**: The result is an anti-derivative of f(x), meaning a function whose derivative is f(x).
- **C**: The constant of integration, since indefinite integrals represent a family of functions differing by a constant.
- **Indefinite Integrals**: Used when we need to recover an original function from its derivative or rate of change (e.g., position from velocity, surplus in economics).

Definite Integral

- The **definite integral** of a function f(x) from a to b is written as

$$\int_a^b f(x)dx = F(b) - F(a).$$

- **f(x)**: The function to be integrated.
- **dx**: The variable of integration.
- **F(b)−F(a)**: The result of the definite integral, which gives the difference between the anti-derivative evaluated at b and at a.
- **Definite Integrals**: Used when we want to find the total value accumulated over an interval (e.g., area, distance, work done, population growth).

Applications of Integration

Some real real-life applications of integrations are

1. Physics: Computing displacement from velocity.

2. Engineering: Calculating the area and volume of irregular shapes.

3. Economics: Determining consumer surplus and producer surplus.

4. **Biology**: Modelling **population growth** over time.

5. **Electrical Engineering**: Finding the total **charge** accumulated from a variable current.

Some Useful Integration Formulas.

- $\int dx = x + C$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
- $\int e^x dx = e^x + C$
- $\int e^{ax} dx = \frac{e^{ax}}{a} + C$
- $\int \frac{1}{x} dx = \ln x + C$
- $\int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + C$
- $\int \frac{f'(x)}{f(x)} dx = \ln f(x) + C$
- $\int \cos x\, dx = \sin x + C$
- $\int \sin x\, dx = -\cos x + C$

Here are some more formulas.

- $\int \cos(ax) dx = \frac{\sin(ax)}{a} + C$
- $\int \sin(ax) dx = \frac{-\cos(ax)}{a} + C$
- $\int \sec^2 x\, dx = \tan x + C$
- $\int \csc^2 x\, dx = -\cot x + C$
- $\int \sec x \tan x\, dx = \sec x + C$
- $\int \csc x \cot x\, dx = -\csc x + C$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$\int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} |x| + C$$

PROPERTIES OF THE INDEFINITE INTEGRAL

Our first properties of antiderivatives follow directly from the simple constant factor, sum, and difference rules for derivatives.

5.2.3 THEOREM *Suppose that $F(x)$ and $G(x)$ are antiderivatives of $f(x)$ and $g(x)$, respectively, and that c is a constant. Then:*

(a) *A constant factor can be moved through an integral sign; that is,*

$$\int cf(x)\,dx = cF(x) + C$$

(b) *An antiderivative of a sum is the sum of the antiderivatives; that is,*

$$\int [f(x) + g(x)]\,dx = F(x) + G(x) + C$$

(c) *An antiderivative of a difference is the difference of the antiderivatives; that is,*

$$\int [f(x) - g(x)]\,dx = F(x) - G(x) + C$$

Rule of Definite 1ntegral

Fundamental Theorem of Calculus If F', the derivative of F, is continuous, then

$$\int_a^b F'(x)\,dx = F(b) - F(a).$$

$x = a$ lower limit of integration
$x = b$ upper limit of integration

Example

Example 3

$$\int (3x^6 - 2x^2 + 7x + 1)\,dx = 3\int x^6\,dx - 2\int x^2\,dx + 7\int x\,dx + \int 1\,dx$$

$$= \frac{3x^7}{7} - \frac{2x^3}{3} + \frac{7x^2}{2} + x + C$$

Examples

Example 4 Evaluate

$$\text{(a)} \int \frac{\cos x}{\sin^2 x}\,dx \qquad \text{(b)} \int \frac{t^2 - 2t^4}{t^4}\,dt \qquad \text{(c)} \int \frac{x^2}{x^2+1}\,dx$$

Solution (a).

$$\int \frac{\cos x}{\sin^2 x}\,dx = \int \frac{1}{\sin x}\frac{\cos x}{\sin x}\,dx = \int \csc x \cot x\,dx \underbrace{=}_{\text{Formula 8 in Table 5.2.1}} -\csc x + C$$

Solution (b).

$$\int \frac{t^2 - 2t^4}{t^4}\,dt = \int \left(\frac{1}{t^2} - 2\right) dt = \int (t^{-2} - 2)\,dt$$

$$= \frac{t^{-1}}{-1} - 2t + C = -\frac{1}{t} - 2t + C$$

Solution (c). By adding and subtracting 1 from the numerator of the integrand, we can rewrite the integral in a form in which Formulas 1 and 12 of Table 5.2.1 can be applied:

$$\int \frac{x^2}{x^2+1}\,dx = \int \left(\frac{x^2+1}{x^2+1} - \frac{1}{x^2+1}\right) dx$$

$$= \int \left(1 - \frac{1}{x^2+1}\right) dx = x - \tan^{-1} x + C$$

Examples

10. World annual natural gas[6] consumption, N, in millions of metric tons of oil equivalent, is approximated by $N = 1770 + 53t$, where t is in years since 1990.

(a) How much natural gas was consumed in 1990? In 2010?

(b) Estimate the total amount of natural gas consumed during the 20-year period from 1990 to 2010.

Examples

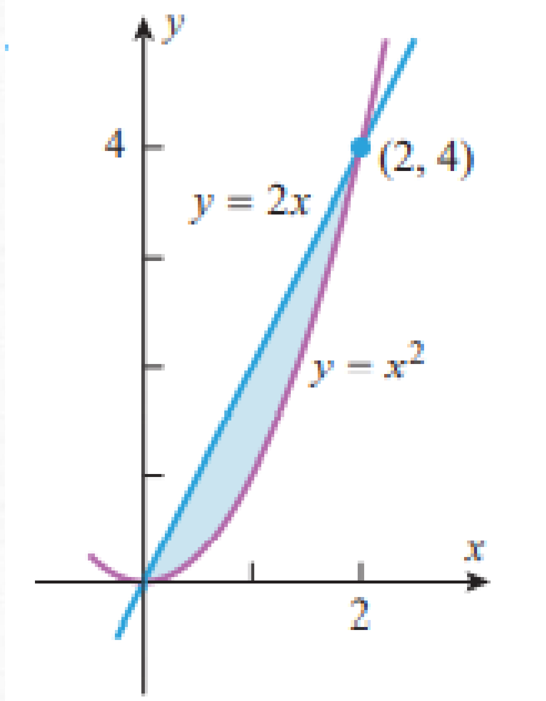
70. A traffic engineer monitors the rate at which cars enter the main highway during the afternoon rush hour. From her data she estimates that between 4:30 P.M. and 5:30 P.M. the rate $R(t)$ at which cars enter the highway is given by the formula $R(t) = 100(1 - 0.0001t^2)$ cars per minute, where t is the time (in minutes) since 4:30 P.M.

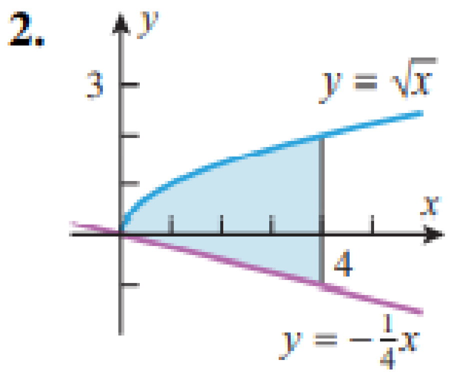
(a) When does the peak traffic flow into the highway occur?

(b) Estimate the number of cars that enter the highway during the rush hour.

Examples

Find the area of the shaded region.





- Thank You